Discrete Event Simulator

Viren Mehta (25M2132)
Shikhar Rai (25M2128)

Features

- Multi-core server machine
- Multi-threaded server
- Round-robin scheduling/FIFO Scheduling
- Closed loop system

Input

- n_users
- n_CPUs
- warmup-time
- maxRuns
- BufferCapacity
- threadpoolSize
- quantum
- ctxSwitchTime
- serviceTimeMean
- thinkTimeMin
- thinkTimeMax

Output

- throughput
- goodput
- badput
- responseTime
- utilization

Correctness of the code

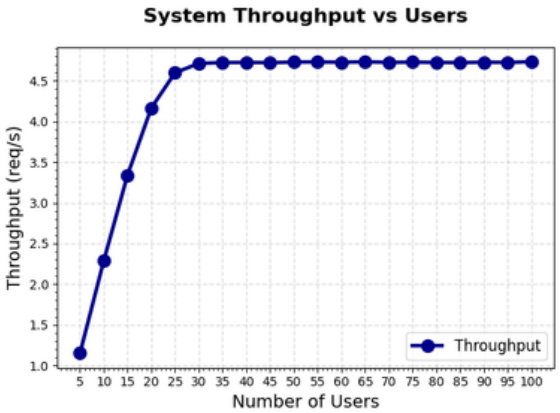
System Characteristics

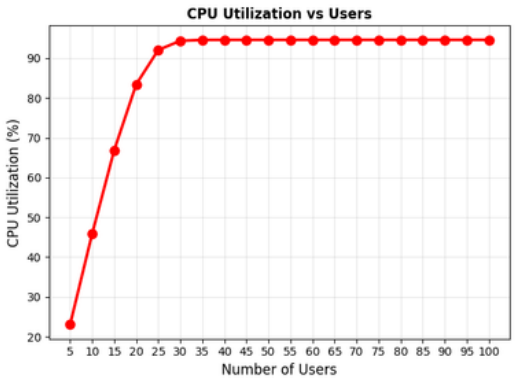
- Multi-core server system with 4 CPU cores
- Multi-threaded web server with thread pool size = 100
- Thread-per-request model: requests are assigned threads until pool is full; beyond that, requests wait in queue
- Finite request buffer (queue size = 500), allowing request drops under high load
- Scheduling policy: Round Robin, with time quantum = 0.2 units and FIFO
- Context switching overhead = 0.01 units, modeled dynamically based on system load
- Different service time distribution with mean = 0.8 units ($\mu \approx 1.25$ req/unit per core)
- User think time follows a uniform distribution [2, 5] units (mean ≈ 3.5 units), ensuring non-zero mode
- Closed-loop workload model: users alternate between thinking and issuing requests
- Request timeout = 5 units; timed-out requests contribute to badput
- Simulation duration = 10000 units, with warmup period = 2000 units to remove transient effects

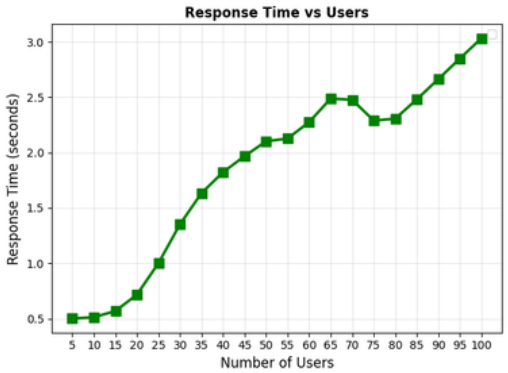
Exponential Service Time Distribution
Round Robin Scheduling Policy

Simulation Parameters - 1 (Number of users)

- n_users = [5, 100]
- n_CPUs = 4
- warmUp-Time = 2000
- maxRuns = 20
- bufferCapacity=50
- threadpoolSize = 100
- quantum = 0.2
- ctxSwitchTime = 0.01
- serviceTimeMean = 0.8
- thinkTimeMin = 2
- thinkTimeMax=5







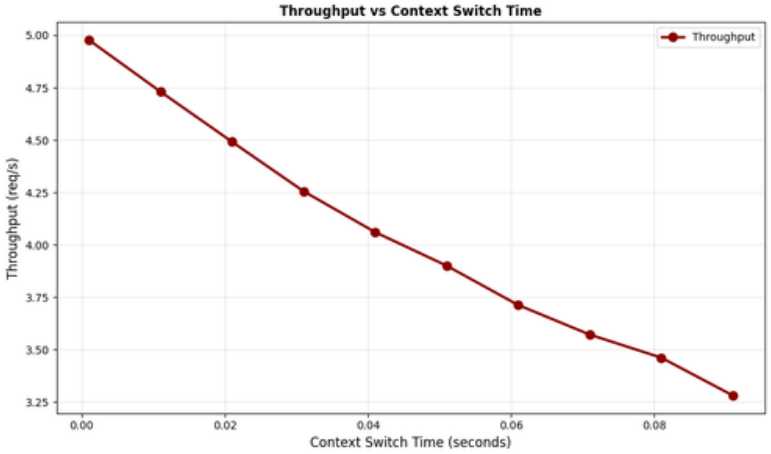
- Throughput increases initially → more load, better utilization
- Saturates around ~25–30 users (~4.7 req/s)
- Remains flat → CPU-bound region

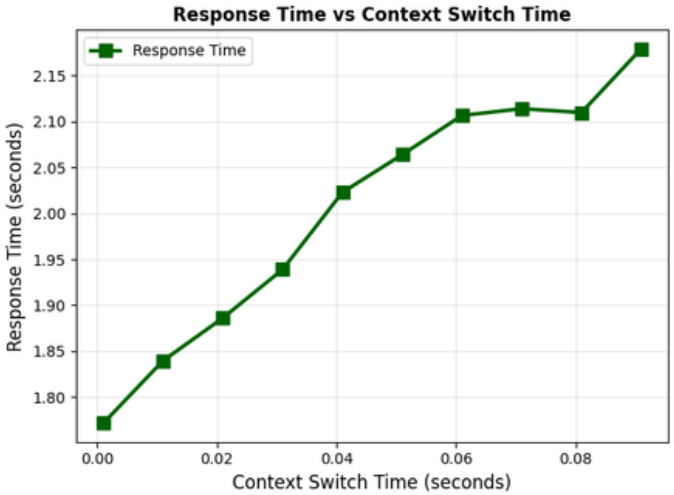
- Rapid increase at low users
- Reaches ~95% around ~25 users
- Stays flat → full CPU utilization

- Gradual increase at low load
- Sharp rise after ~30 users → queue buildup
- Fluctuations due to EXP variability + RR scheduling
- Continues increasing under high load

Simulation Parameters - 2 (Context Switching Time)

- n_users = 50
- n_CPUs = 4
- warmupTime= 2000
- maxRuns = 20
- bufferCapacity = 50
- threadpoolSize = 100
- quantum = 0.2
- ctxSwitchTime = [0.001, 0.1, 0.01]
  serviceTimeMean = 0.8
- thinkTimeMin = 2
- thinkTimeMax=5



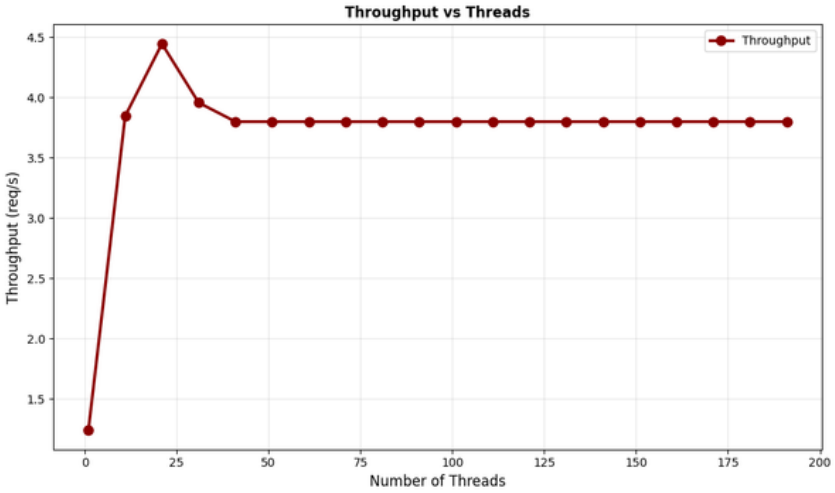


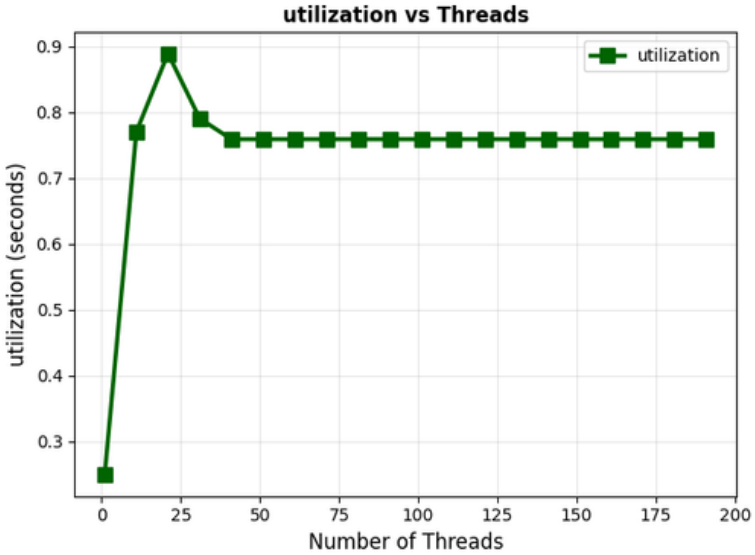
- Throughput decreases steadily as context switch time increases
- RR causes frequent preemptions, amplifying overhead

- Response time increases gradually with context switch time
- Increase driven by overhead + queueing delays

Simulation Parameters - 3 (Threadpool)

- n_users = 25
- n_CPUs = 4
- warmUp-Time = 2000
- maxRuns = 20
- bufferCapacity = 100
- threadpoolSize = [1,200,10]
- quantum=0.2
- ctxSwitchTime = 0.01
- serviceTimeMean = 0.8
- thinkTimeMin = 2
- thinkTimeMax=5





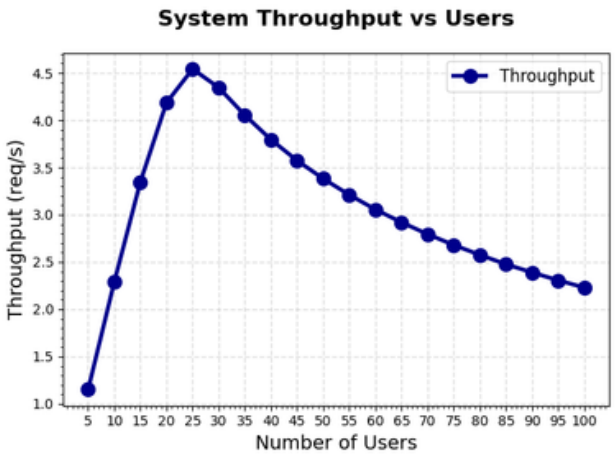
- Sharp rise at low threads → thread-limited
- Peaks around ~15–20 threads
- Then drops and stabilizes lower
- Drop due to EXP variability + RR overhead

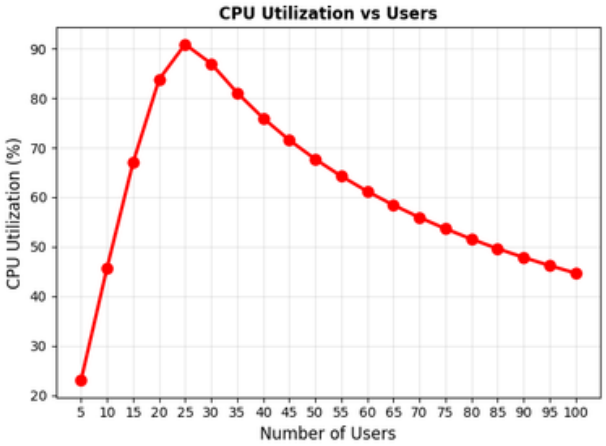
- Low threads → underutilized CPU
- Peaks around ~90%
- Then stabilizes lower (~75–78%)
- Lower due to frequent context switching + uneven job sizes

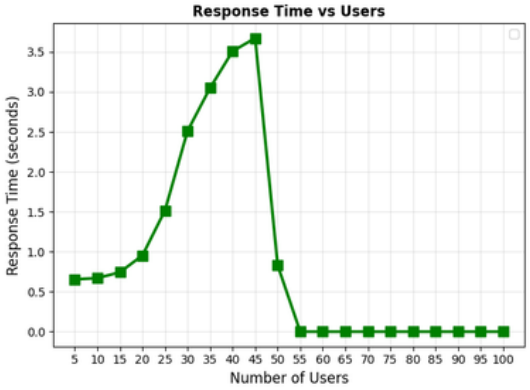
Uniform Service Time Distribution
Round Robin Scheduling Policy

Simulation Parameters - 1 (Number of users)

- n_users = [5, 100]
- n_CPUs = 4
- warmUp-Time = 2000
- maxRuns = 20
- bufferCapacity=50
- threadpoolSize = 100
- quantum = 0.2
- ctxSwitchTime = 0.01
- serviceTimeMean = 0.8
- thinkTimeMin = 2
- thinkTimeMax=5







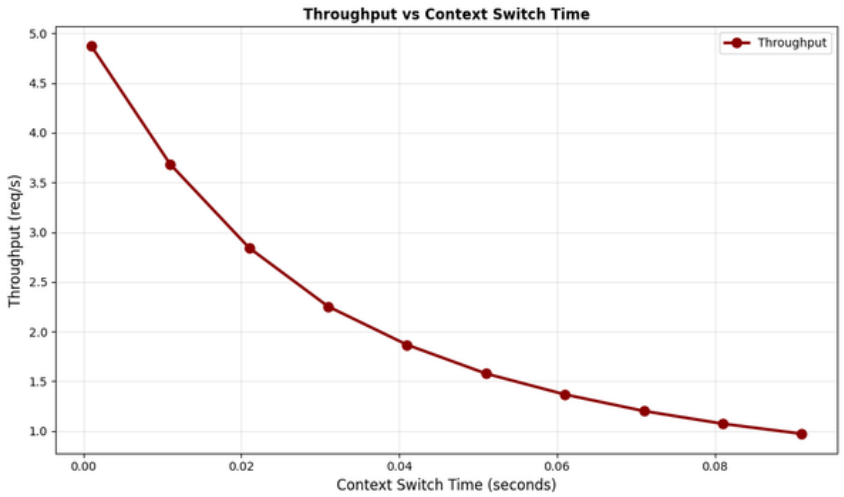
- Throughput rises initially → better utilization
- Peaks around ~25 users
- Then declines steadily → system overload
- Drop due to RR overhead + increasing queueing

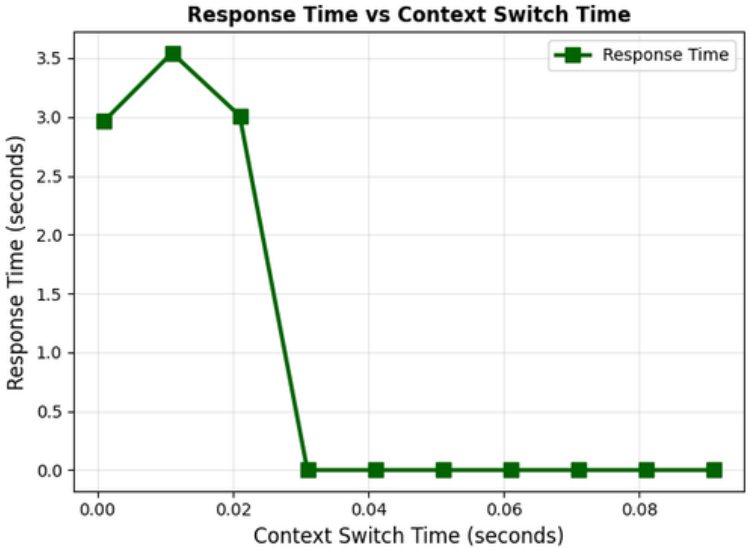
- Increases rapidly → peaks ~90%
- Then declines gradually
- Drop due to overhead + inefficient scheduling under load

- Increases steadily → queue buildup
- Sharp rise near ~40–50 users → saturation
- Sudden drop to ~0 (due to goodput → 0)

Simulation Parameters - 2 (Context Switching Time)

- n_users = 50
- n_CPUs = 4
- warmupTime= 2000
- maxRuns = 20
- bufferCapacity = 50
- threadpoolSize = 100
- quantum = 0.2
- ctxSwitchTime = [0.001, 0.1, 0.01]
- serviceTimeMean = 0.8
- thinkTimeMin = 2
- thinkTimeMax=5



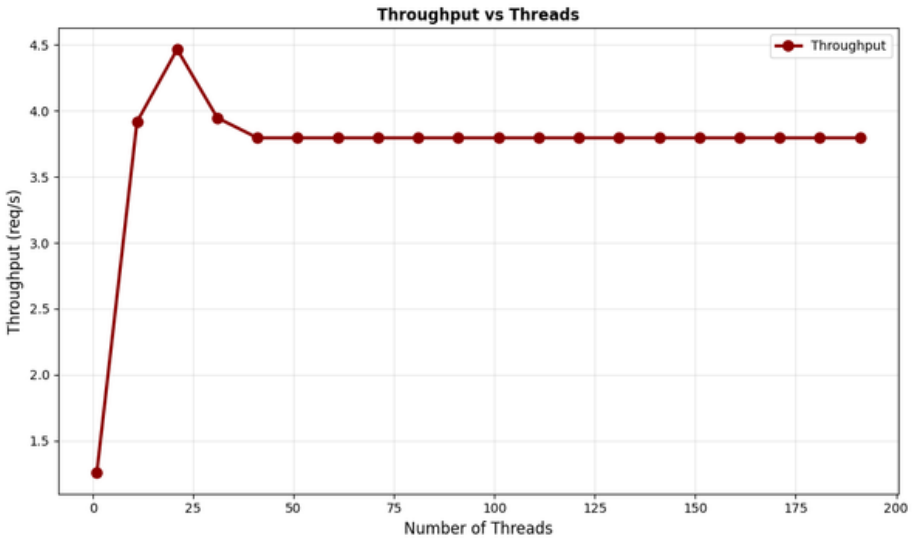


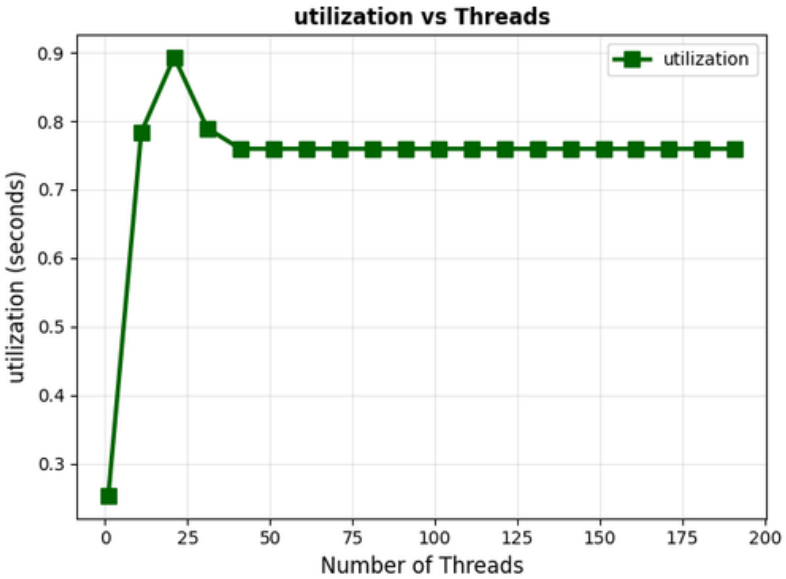
- Throughput decreases steadily as context switch time increases
- RR incurs frequent context switches, amplifying overhead

- Initially high due to RR slicing + overhead
- Drops sharply to ~0 (due to goodput → 0)
- Indicates system fails to complete requests at high overhead

Simulation Parameters - 3 (Threadpool)

- n_users = 25
- n_CPUs = 4
- warmUp-Time = 2000
- maxRuns = 20
- bufferCapacity = 100
- threadpoolSize = [1,200,10]
- quantum=0.2
- ctxSwitchTime = 0.01
- serviceTimeMean = 0.8
- thinkTimeMin = 2
- thinkTimeMax=5





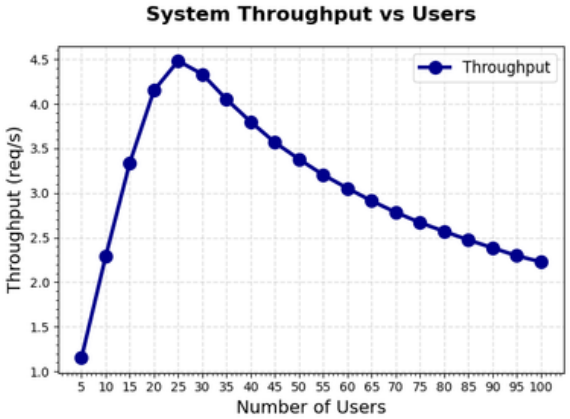
- Sharp rise at low threads → thread-limited
- Peaks around ~15–20 threads
- Then drops slightly and stabilizes lower than FIFO
- Drop due to context switching overhead (RR)

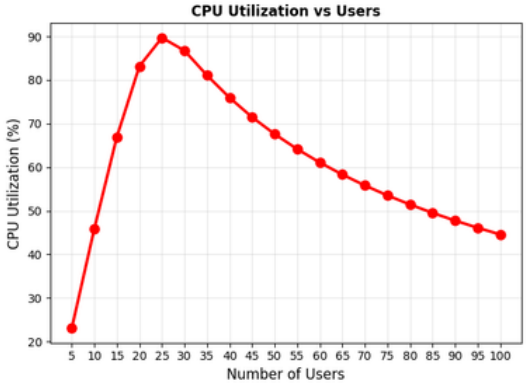
- Low threads → underutilized CPU
- Rapid rise to ~95–98%
- Then stabilizes slightly lower (~94–95%)
- Drop due to long jobs blocking (FIFO + EXP variance)

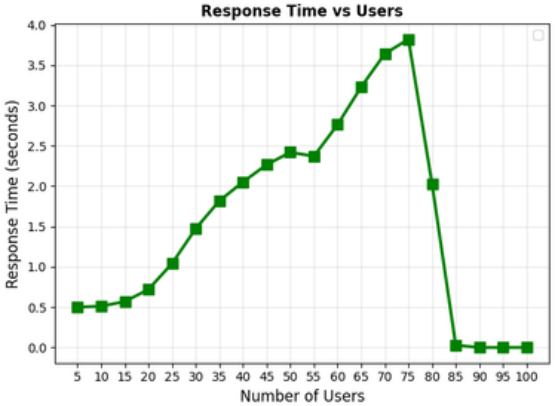
Exponential Service Time Distribution
FIFO Scheduling Policy

Simulation Parameters - 1 (Number of users)

- n_users = [5, 100]
- n_CPUs = 4
- warmUp-Time = 2000
- maxRuns = 20
- bufferCapacity=50
- threadpoolSize = 100
- quantum = 0.2
- ctxSwitchTime = 0.01
- serviceTimeMean = 0.8
- thinkTimeMin = 2
- thinkTimeMax=5







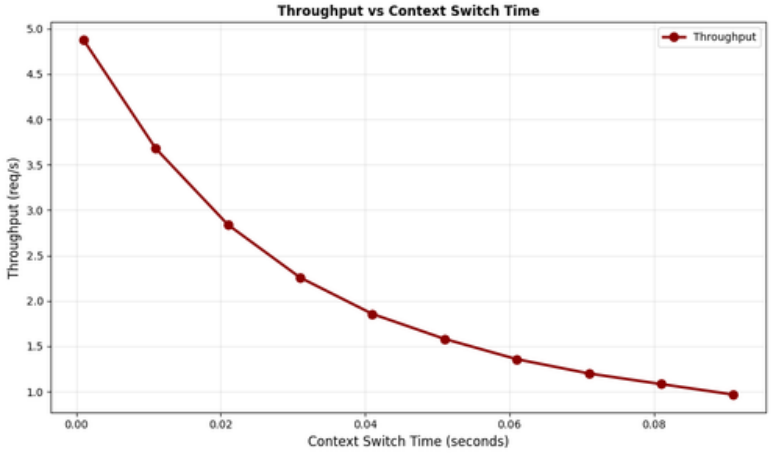
- Throughput increases initially → better utilization
- Peaks around ~20–25 users
- Then declines steadily → overload
- Drop due to EXP variability + FIFO blocking

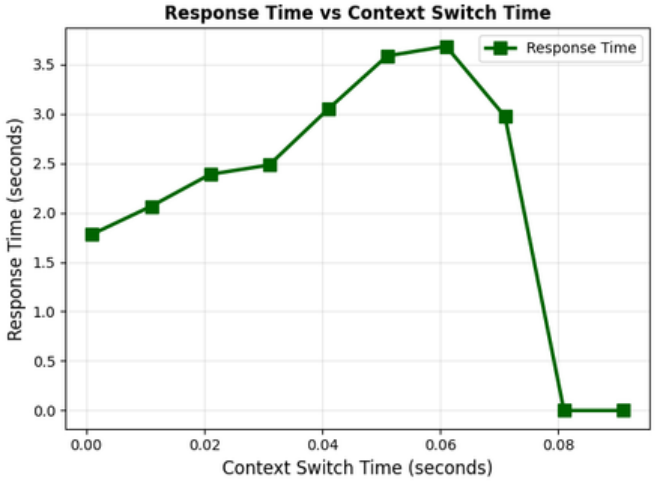
- Rapid rise → peaks ~90%
- Then declines gradually
- Drop due to inefficiency from long jobs blocking queue

- Gradual increase at low load
- Sharp rise after ~30 users → queue buildup + blocking
- Peaks very high (~3.8 sec)
- Sudden drop to ~0 (artifact due to goodput → 0)

Simulation Parameters - 2 (Context Switching Time)

- n_users = 50
- n_CPUs = 4
- warmupTime= 2000
- maxRuns = 20
- bufferCapacity = 50
- threadpoolSize = 100
- quantum = 0.2
- ctxSwitchTime = [0.001, 0.1, 0.01]
- serviceTimeMean = 0.8
- thinkTimeMin = 2
- thinkTimeMax=5



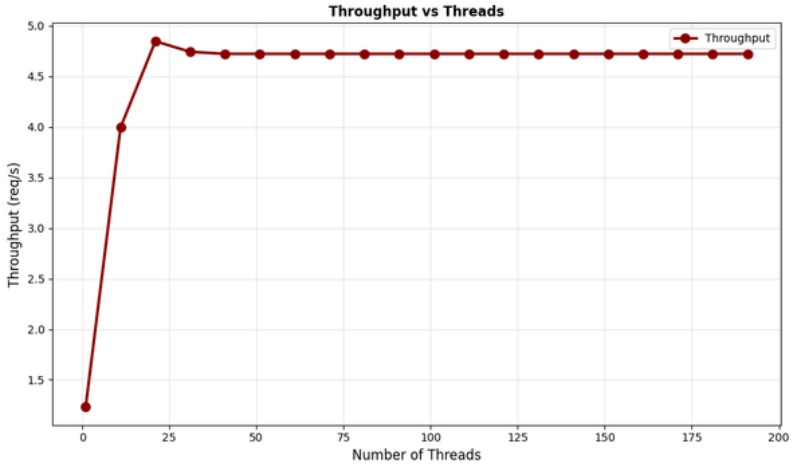


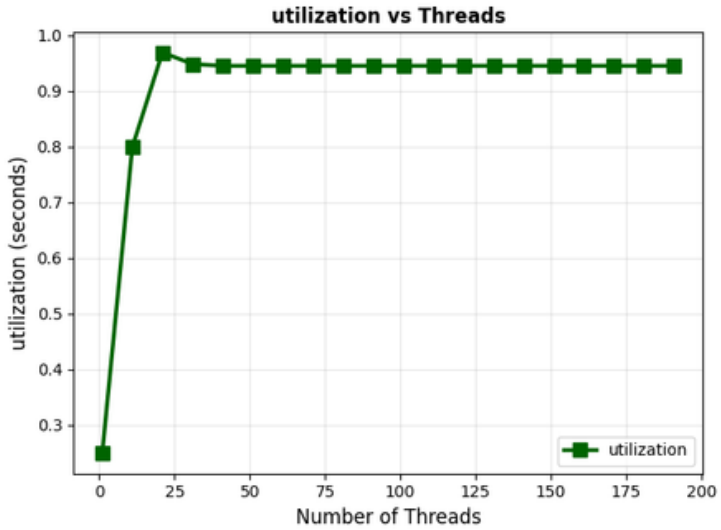
- Throughput decreases monotonically as context switch time increases
- CPU spends more time in overhead than useful work

- Response time increases steadily → longer waiting + execution delays
- Peaks around moderate context switch → system nearing saturation
- Sudden drop to ~0 (due to goodput → 0)

Simulation Parameters - 3 (Threadpool)

- n_users = 25
- n_CPUs = 4
- warmUp-Time = 2000
- maxRuns = 20
- bufferCapacity = 100
- threadpoolSize = [1,200,10]
- quantum=0.2
- ctxSwitchTime = 0.01
- serviceTimeMean = 0.8
- thinkTimeMin = 2
- thinkTimeMax=5





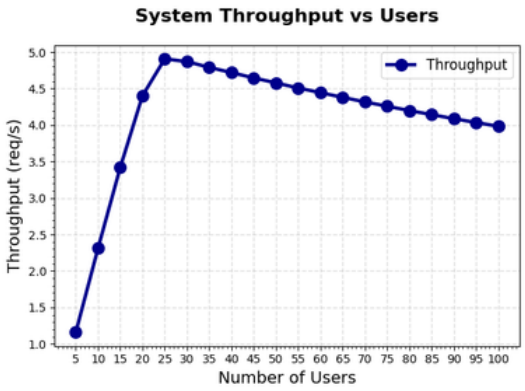
- Sharp rise at low threads → thread-limited region
- Peaks around ~15–20 threads
- Then slightly drops and stabilizes
- Drop due to higher variability (EXP) causing inefficiencies

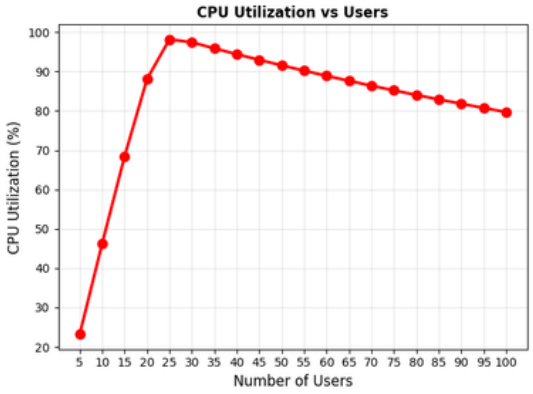
- Low threads → underutilized CPU
- Rapid rise to ~95–98%
- Then stabilizes slightly lower (~94–95%)
- Drop due to long jobs blocking (FIFO + EXP variance)

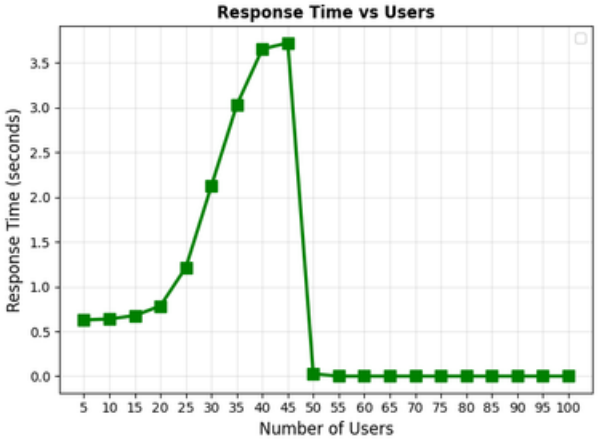
Uniform Service Time Distribution
FIFO Scheduling Policy

Simulation Parameters - 1 (Number of users)

- n_users = [5, 100]
- n_CPUs = 4
- warmUp-Time = 2000
- maxRuns = 20
- bufferCapacity=50
- threadpoolSize = 100
- quantum = 0.2
- ctxSwitchTime = 0.01
- serviceTimeMean = 0.8
- thinkTimeMin = 2
- thinkTimeMax=5







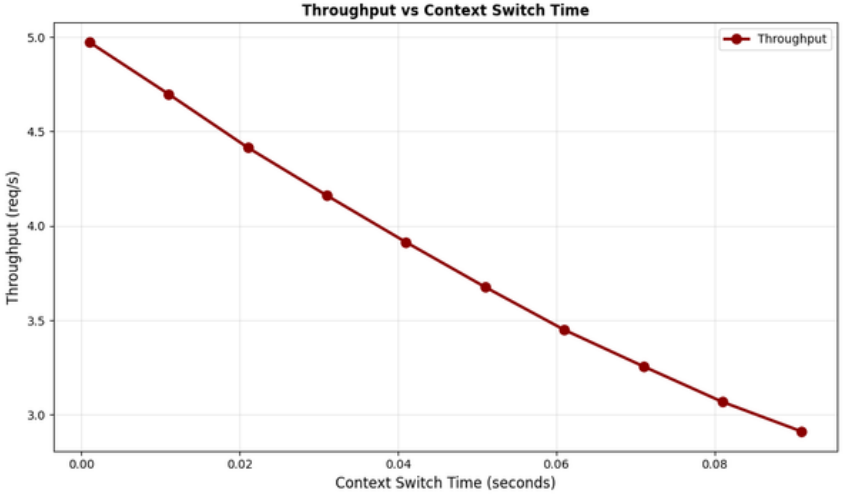
- Throughput increases initially → better utilization
- Peaks around ~20–25 users
- Then gradually declines → overload
- Decline smoother than EXP due to lower variability (uniform)

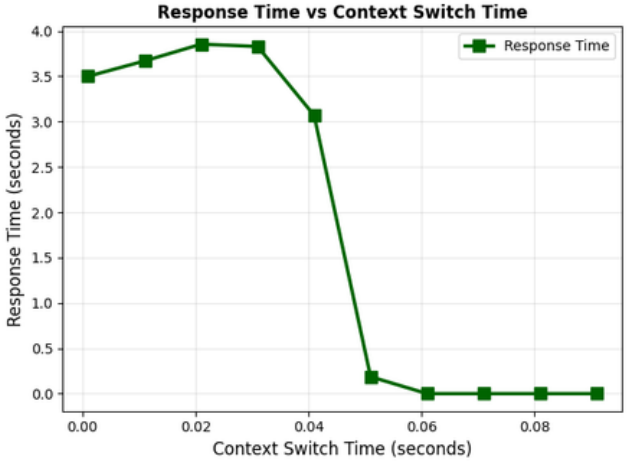
- Rapid rise → peaks near ~100%
- Then slow decline
- Drop due to queueing inefficiency and FIFO blocking

- Gradual increase at low load
- Sharp rise after ~30–40 users → queue buildup
- Peaks around ~3.5–3.7 sec
- Sudden drop to ~0 (due to goodput → 0)

Simulation Parameters - 2 (Context Switching Time)

- n_users = 50
- n_CPUs = 4
- warmupTime= 2000
- maxRuns = 20
- bufferCapacity = 50
- threadpoolSize = 100
- quantum = 0.2
- ctxSwitchTime = [0.001, 0.1, 0.01]
- serviceTimeMean = 0.8
- thinkTimeMin = 2
- thinkTimeMax=5



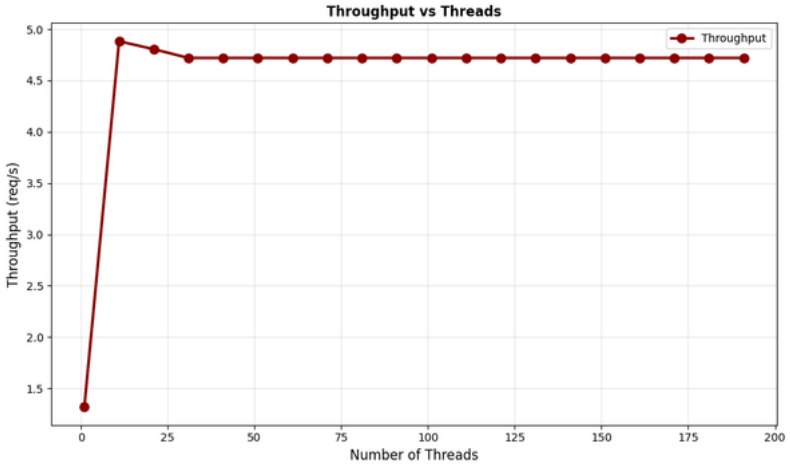


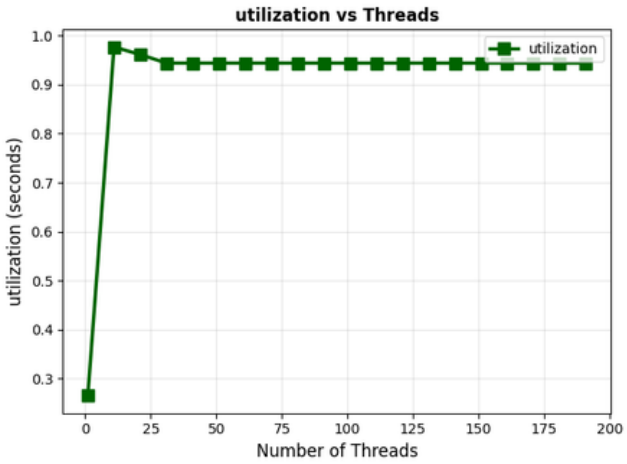
- Throughput decreases steadily as context switch time increases
- Nearly linear decline → more time spent in overhead

- Response time slightly increases at first → more overhead
- Peaks around moderate context switch
- Then suddenly drops to ~0 (due to goodput → 0)

Simulation Parameters - 3 (Threadpool)

- n_users = 25 n_CPUs = 4
- warmUp-Time = 2000
- maxRuns = 20
- bufferCapacity = 100
- threadpoolSize = [1,200,10]
- quantum=0.2
- ctxSwitchTime = 0.01
- serviceTimeMean = 0.8
- thinkTimeMin = 2
- thinkTimeMax=5



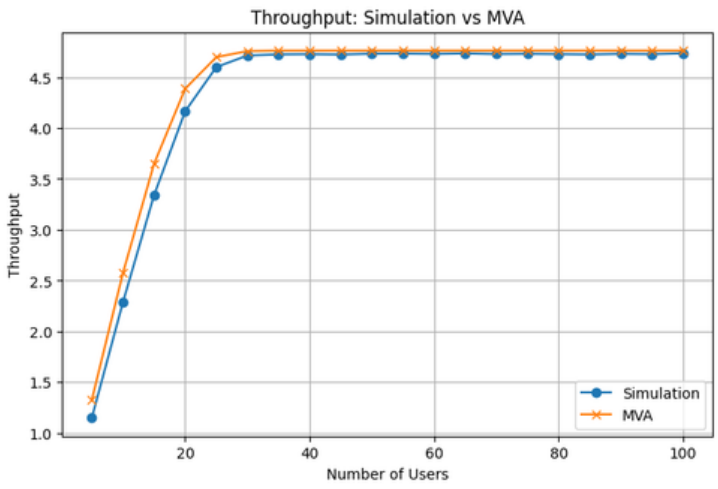


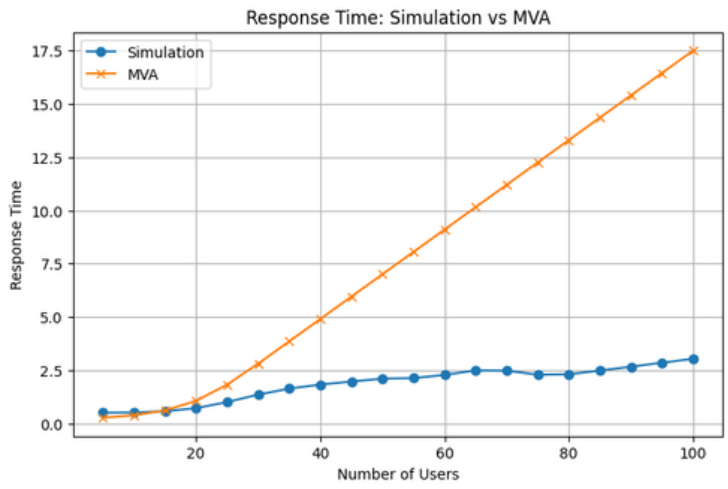
- Throughput rises quickly at low threads → system is thread-limited
- Peaks around ~10–20 threads
- Then plateaus → system becomes CPU-bound
- Extra threads only increase waiting, not throughput

- Low threads → underutilized CPU
- Rapid increase as threads grow
- Stabilizes ~95% → full utilization reached
- Slight drop due to overhead + FIFO inefficiency

Mean Value Analysis
Vs
Simulated Model

Exponential And RoundRobin Config

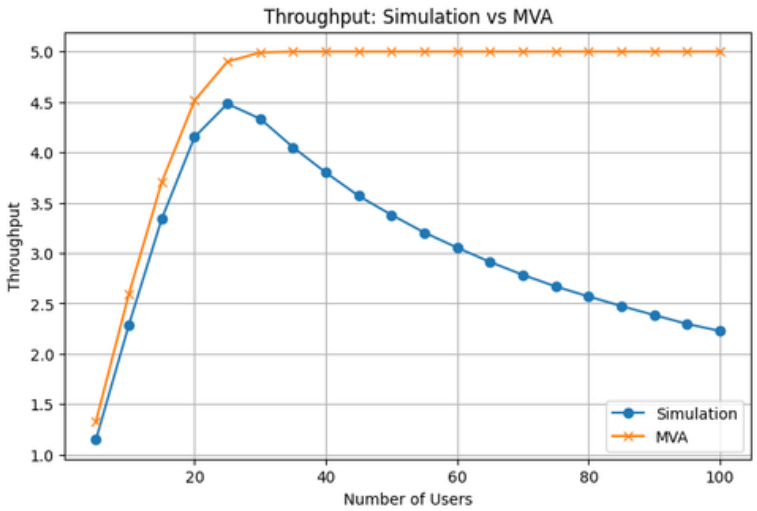




- Both curves closely match across all users
- Throughput saturates around system capacity (~4.7 req/s)
- MVA slightly overestimates at low users
- Throughput is capacity-driven, so MVA predicts it well

- Simulation increases gradually → realistic queueing behavior
- MVA increases sharply → overestimates delay at high load
- Large gap at higher users
- MVA fails to capture RR scheduling and system dynamics

Exponential and FIFO Config



- MVA shows constant saturation (~5 req/s)
- Simulation peaks early and then declines with more users
- Drop due to FIFO blocking + queue buildup
- MVA ignores blocking → overestimates throughput

- Simulation increases initially → queueing delay
- Then drops sharply (artifact due to goodput collapse)
- MVA increases steadily → assumes smooth behavior
- FIFO + EXP causes severe blocking not captured by MVA

Conclusion and Unanswered Questions

- Throughput is well predicted by MVA across all configurations, as it primarily depends on system capacity (cores and service rate)
- Response time shows major deviations from MVA, especially at higher loads, due to:
  - Service time variability (Exponential distribution)
  - Scheduling overhead (Round Robin)
  - Head-of-line blocking (FIFO)
  - Finite buffers and thread limits
  - Timeout mechanisms
- Improper graph behaviors:
  - Sudden drop in response time at high load occurs when:
    - Goodput → 0 (no successful completions)
    - Response time becomes undefined but appears low

End